

Digital Signal Analysis

Quiz, 2026

Answer all the questions, each question carry equal marks. Time: 45 minutes

1. Let $x[n] = \{2, 3, 2, 3, 2, 1\}$ then three DFTs $X_1[k]$, $X_2[k]$ and $X_3[k]$ are computed considering $N = 8, 6$ and 4 respectively. Using the equation $\frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi k n}{N}}$, $x_1[n]$, $x_2[n]$ and $x_3[n]$ are reconstructed from $X_1[k]$, $X_2[k]$ and $X_3[k]$ respectively. Compute $x_1[n]$, $x_2[n]$ and $x_3[n]$. [3 Marks]

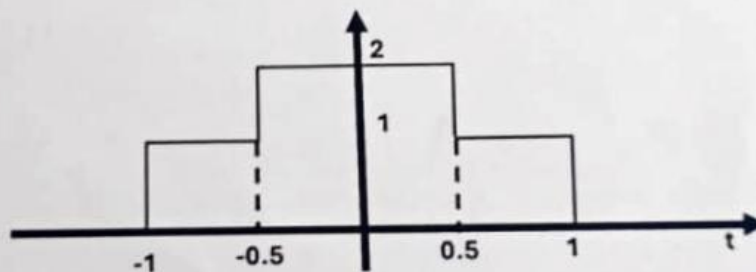
2. Let $x[n] = \{2, 0, 2\}$ and its $X[\omega] = 4 \cos \omega$ then find its DFT $X[k]$ for $N = 8$? [2 Marks]

3. Let $x[n] = \{2, 0, -2, 1, 2, 1\}$ then find $X[\omega]$? [1 Mark]

4. Let $x_1[n] = \{2, 3, 2, 3, 2, 1\}$ and $x_2[n] = \{2, 3, 2, 3, 2, 1, 0, 0\}$ then DFTs $X_1[k]$ and $X_2[k]$ are computed using $N = 8$ for $x_1[n]$ and $x_2[n]$ respectively. Can the $X_1[k]$ and $X_2[k]$ are the same justify your answer? [2 Marks]

5. For $x[n] = u[n] - u[n-1]$ then find $X[\omega]$? [1 Mark]

6. Let's consider $x(t)$ as shown below then find $x[n]$ with a sampling frequency of 1.5Hz . [2 Marks]



7. Is the following signal periodic $\cos(5n + \frac{\pi}{6})$? Please justify your answer. [2 Marks]

8. Compute the Nyquist rate of the signal $x(t) = \sin(360\pi t)$? [1 Mark]

Useful Formulas:

$$\text{DTFT: } X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$\text{DFT: } X[k] = \sum_{n=0}^{L-1} x[n] e^{-j \frac{2\pi k n}{N}}$$

$$\text{IDFT: } x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi k n}{N}}$$